

CLASS 03 / SEP 8

3D PRINTING

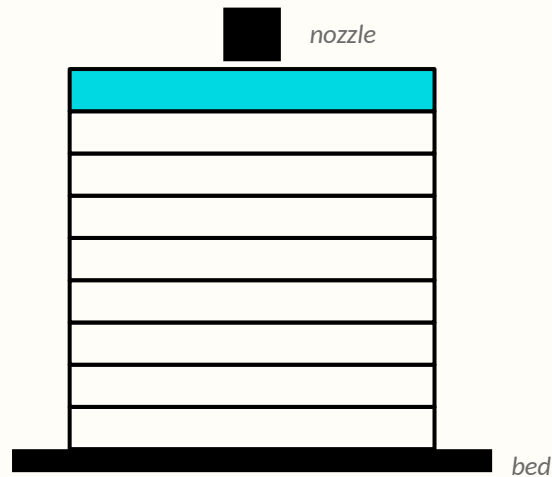
GEOMETRY THAT SURVIVES CONTACT WITH A MACHINE

MOSTLY A WORKING SESSION / BRING YOUR MODEL

WHAT THE MACHINE DOES

One thing, repeatedly. Everything else follows from this.

A NOZZLE DRAWS ONE FLAT LAYER,
THE BED DROPS, IT DRAWS THE NEXT.



IT BUILDS UPWARD, NEVER SIDeways

Material can only be placed on material that is already there.

LAYERS ARE THE WEAK DIRECTION

A print snaps between layers long before it snaps across one.

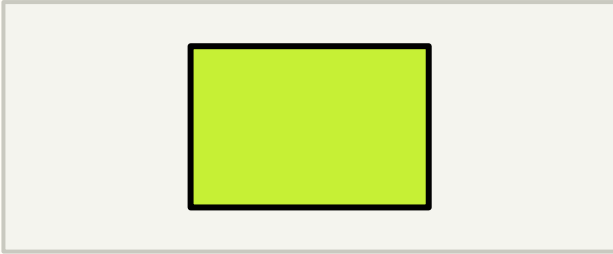
EVERY DECISION IS A TIME COST

Finer layers, more material, more supports — all of it is hours on a shared machine.

MANIFOLD OR IT DOES NOT PRINT

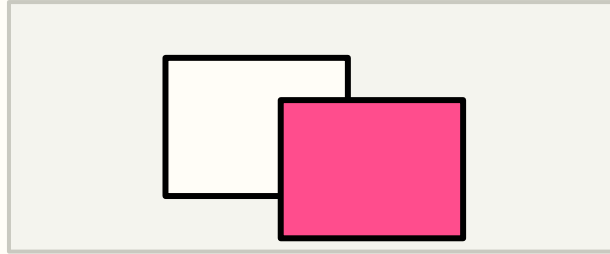
A model can look perfect on screen and be meaningless to a slicer.

WATERTIGHT



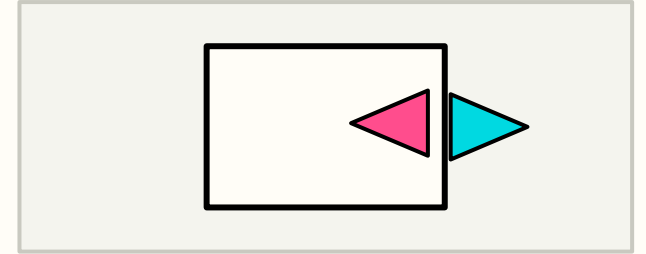
No gaps. Every edge is shared by exactly two faces. If water could leak out, it is not a solid.

NO SELF-INTERSECTION



A shape that passes through itself has no unambiguous inside. The slicer cannot decide what to fill.

NORMALS OUTWARD



Every face has a front and a back. A flipped face tells the slicer that outside is inside.

CHECK BEFORE YOU SLICE. A BOOLEAN THAT LEFT AN OPEN EDGE IS THE MOST COMMON CAUSE OF A FAILED FILE.

OVERHANGS

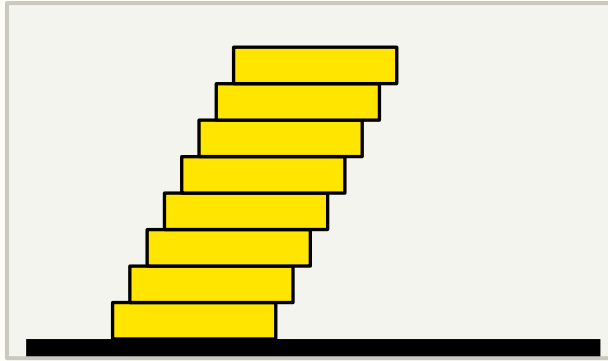
Each layer needs something under it. How much is the whole question.

VERTICAL



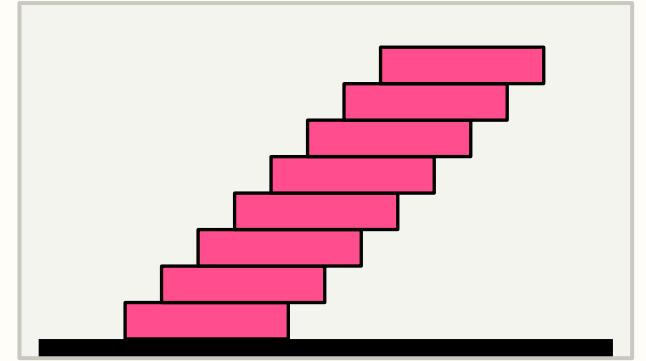
Every layer sits fully on the one below. Perfect.

45°



Half of each layer is supported. This is about the limit.

STEEP

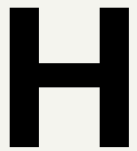


Most of the layer hangs in air. It will droop or fail.

THE 45° RULE: BEYOND ABOUT 45° FROM VERTICAL, A SURFACE NEEDS SUPPORT — OR A DIFFERENT ORIENTATION.

THE LETTER TEST

Four letters, four printing conditions. Find yours in your own model.

A large, bold, black letter 'H' is centered within a light gray square. Below the square is a thick black horizontal bar.

PRINTS FINE

BRIDGE

The crossbar spans between two legs that are already there. Anchored at both ends, so it can stretch across.

A large, bold, black letter 'V' is centered within a light gray square. Below the square is a thick black horizontal bar.

FINE TO 45°

SLOPE

Each layer steps outward a little. Gentle is fine; past about 45° from vertical it starts to droop.

A large, bold, black letter 'T' is centered within a light gray square. Below the square is a thick black horizontal bar.

DROOPS

CANTILEVER

The arms start in mid-air and are anchored at one end only. Nothing to stretch to.

A large, bold, black letter 'i' is centered within a light gray square. Below the square is a thick black horizontal bar.

IMPOSSIBLE

ISLAND

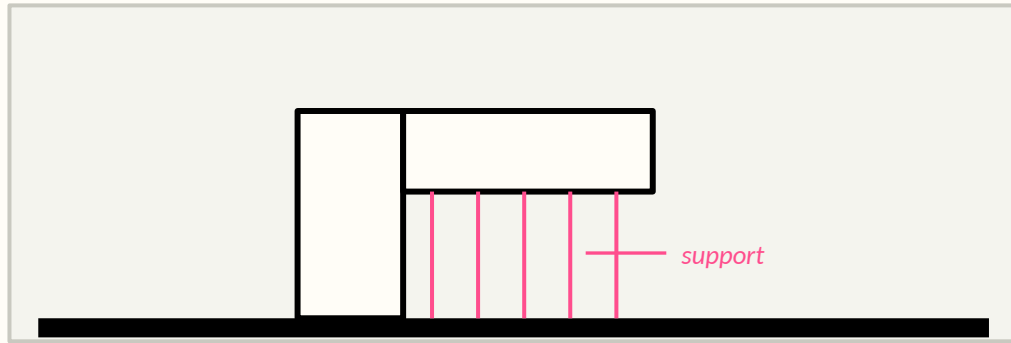
The dot begins with nothing underneath it at all. There is no version of this without support.

A BRIDGE IS ANCHORED AT BOTH ENDS. A CANTILEVER IS NOT. AN ISLAND IS ANCHORED TO NOTHING. THAT IS THE WHOLE DIFFERENCE.

ORIENTATION IS YOUR ONLY LEVER

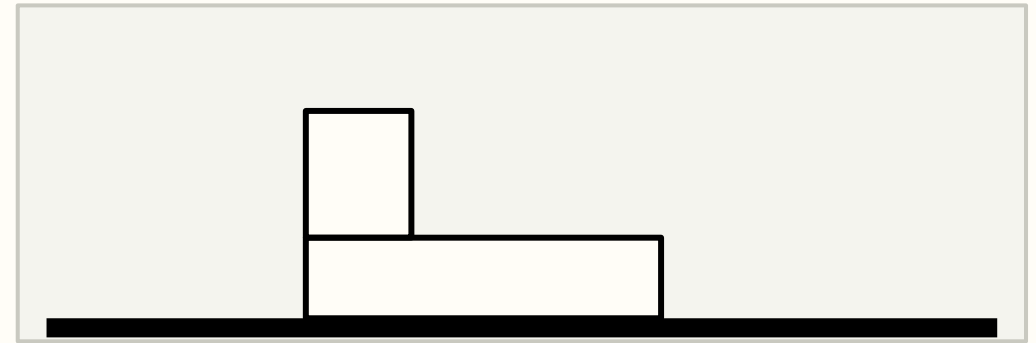
Same geometry. Two ways to place it. Very different prints.

AS MODELLED



Overhang under the arm. Needs support, which costs material, time, and a scarred surface where it is removed.

ROTATED



The same arm now sits on the bed. No support, faster, cleaner face — and stronger, because the load is not across a layer line.

NO SUPPORTS ARE PERMITTED ON THE 0.1 PRINT. IF IT CANNOT PRINT UNSUPPORTED IN ANY ORIENTATION, CHANGE THE GEOMETRY.

WHAT THE SLICER DOES

Turns a solid into a list of moves. This is the same idea as Module 2.

01

CUT INTO LAYERS

The solid is intersected by a stack of horizontal planes.

02

OUTLINE EACH LAYER

Each intersection becomes a closed curve — the perimeter.

03

FILL THE INSIDE

Infill pattern and density. Usually 10–20%, not solid.

04

ORDER THE MOVES

Perimeters, infill, travel. Written out as G-code.

SOUND FAMILIAR?

Intersect a solid with a stack of planes.
Keep the profiles. Order them.

That is Module 2 — and it is what a
slicer has been doing all along.

LIVE NOW — we slice one object together and read the preview: layer count, print time, material, where the supports would go.

FOUR SETTINGS

Everything else can stay at default for this print.

LAYER HEIGHT

0.2–0.3 mm

Finer looks better and takes proportionally longer. Draft is fine for 0.1.

INFILL

10–40%

Prints are hollow with a lattice inside. Solid is almost never needed.

WALLS / PERIMETERS

2–3

This, not infill, is what makes a print feel solid.

SUPPORTS

OFF

Not permitted for 0.1. Solve it with orientation.

PRINT IT OR CUT IT

Printing is not the default. It is the expensive option.

CUT IT

- ▣ Prismatic — one profile, extruded
- ▣ Flat, or made of flat pieces
- ▣ Repeated many times
- ▣ Larger than your hand
- ▣ Site model massing: stacked chipboard

Minutes per sheet. Twenty buildings at once.

PRINT IT

- ▣ Doubly curved — a vault, a shell
- ▣ Undercuts a blade cannot reach
- ▣ Detail in all three axes at once
- ▣ Small and intricate
- ▣ One-off with no flat description

Hours per object. One at a time.

THE TEST: can the object be described as a stack of flat profiles, or one profile extruded? Then cut it. A simple box printed is a box you waited six hours for.

FABLAB — HOW THIS WORKS HERE

Machines are shared. The rules exist so everyone gets a print.

ORIENTATION COMPLETE

No Safety Orientation, no machine access. This was due at the end of Class 2.

NEVER TOUCH THE HOT END

The nozzle runs around 200–250°C and does not look hot. The bed is hot too.

DO NOT RESTART SOMEONE'S PRINT

If a job looks like it is failing, tell staff. Do not intervene.

QUEUE IS FIRST COME

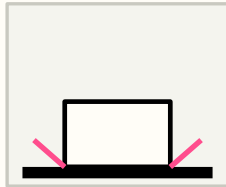
A print that finishes at 3am still finishes. One that starts at the deadline does not.

`[CONFIRM WITH DIEGO: BOOKING METHOD, FILE NAMING, PICK-UP, MATERIAL ALLOCATION]`

SIX WAYS IT GOES WRONG

Learn to name the failure. Naming it is most of the diagnosis.

WARPING



Corners lift off the bed and the base curves.

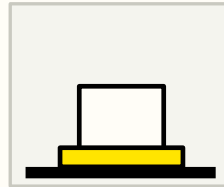
CAUSE

Cooling shrinkage beats bed adhesion.

FIX

Clean bed, add a brim, keep drafts off it.

ELEPHANT'S FOOT



The bottom few layers bulge out wider than the rest.

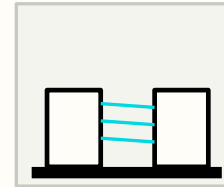
CAUSE

Nozzle too close, or bed too hot.

FIX

Slicer setting — or chamfer the base yourself.

STRINGING



Fine hairs between separate parts of the print.

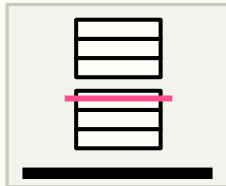
CAUSE

The nozzle oozes while travelling.

FIX

Retraction on. Fewer travel moves — ordering again.

LAYER SEPARATION



The print splits cleanly along a layer line.

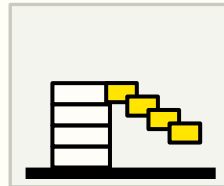
CAUSE

Weak bond, or the load runs across layers.

FIX

Reorient so stress runs along layers, not across.

DROOP



An overhang sags into the space below it.

CAUSE

Nothing underneath to build on.

FIX

Orientation. If that fails, change the geometry.

SPAGHETTI



The nozzle draws in mid-air. A nest of filament.

CAUSE

The part came off the bed part-way through.

FIX

Adhesion. Check the first layer before walking away.

FOR 0.1

YOUR PRINT

PRESENTED AT THE MIDTERM — CLASS 8, OCT 13

60 mm

maximum bounding box

NO SUPPORTS

orientation is the only lever

DRAFT

layer height — this is a test, not a showpiece

1 DIAGRAM

chosen orientation, the overhangs it produces, why not the alternative

Assessed on printability and on your reasoning. Not on craft — that is 3.2's job.

NOW — PREPARE YOUR PRINT

Rest of the session: get a sliced file into the queue with help in the room.



Model checked — manifold, watertight, normals consistent



Scaled to fit within 60 mm



Orientation chosen and justified — no overhang past $\sim 45^\circ$



Sliced. Print time and material noted.



File named and submitted to the queue

LEAVE WITH A FILE IN THE QUEUE.

FIND A TILED FAÇADE

Due next class, before anything is modeled. Look at what the building is made of, not at the building.

WHAT COUNTS

A façade made of one repeated part — a panel, a brick, a shingle, a cast block, a perforated plate. Look at it the way you look at a brick wall and see the brick.

WHAT DOESN'T

A curtain wall of plain glass. A one-off sculptural surface. Anything where you cannot point at the part and say: this, again and again.

BRING

A flat-on elevation photograph or drawing. The repeated part outlined on it. Its width, height and depth in millimetres.

BE READY TO SAY

What the part is. What rule places the copies — stacked, offset, rotated, coursed.

YOU ARE NOT TRACING IT. NEXT WEEK YOU DESIGN YOUR OWN TILE, DRIVEN BY THIS BUILDING.

A short approved list goes out tonight. Propose your own for approval if it is not on it.

ARRAYS BEGIN HERE

The second half of today, and the whole of next week. Same lesson as the slicer.

A SLICER MAKES A LIST

You just watched a solid become a stack of profiles, ordered, then written out as moves.

A list of positions, computed by a rule, executed in order. That is an array, and it is the whole of Module 1.

INDEX → POSITION

$i = 0 \rightarrow p = \text{start}$

$i = 1 \rightarrow p = \text{start} + \text{step}$

$i = 2 \rightarrow p = \text{start} + 2 \times \text{step}$

$p[i] = \text{start} + i \times \text{step}$

AN ARRAY IS NOT A COMMAND THAT DUPLICATES THINGS. IT IS A LIST YOU COMPUTE, THEN PLACE GEOMETRY AT.

BEFORE NEXT CLASS

WATCH

Tutorial 1.1 — points, lists, loops and conditionals. Cartesian, radial, hexagonal, along a curve. Sent tonight.

BRING

Your tiled façade precedent — flat-on image, the repeated part outlined, dimensions. We design your tile from it next class.

FINISH

0.1 is due next class: the axonometric sequence, the written procedure, the precedent analysis.

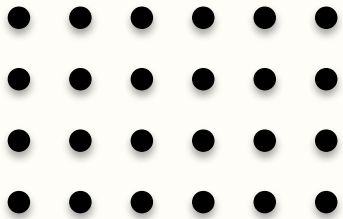
CHECK

Your print is in the queue. It is presented at the Midterm, not next week — but the file has to exist.

FOUR WAYS TO GENERATE THE LIST

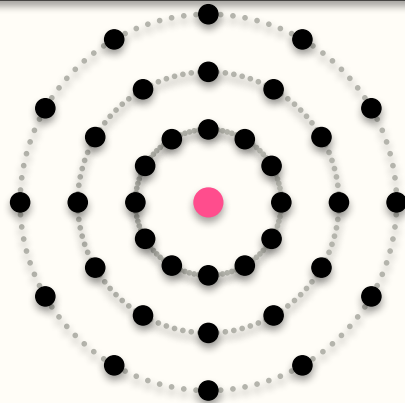
Named today so you can recognize them in a building. Built next week.

CARTESIAN



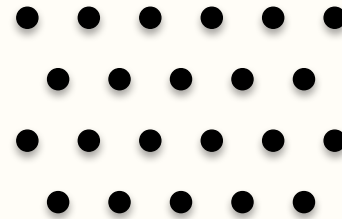
two counters, i and j

RADIAL



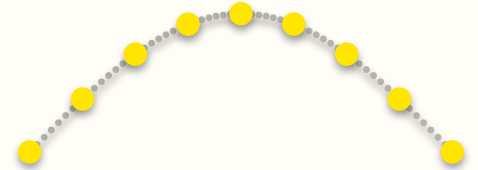
angle and radius

HEXAGONAL



odd rows offset by $dx/2$

ON CURVE



sampled at $t = i/n$

Every one of these is the same operation — move, repeated, with the amount driven by a counter. Only the rule for the counter changes.